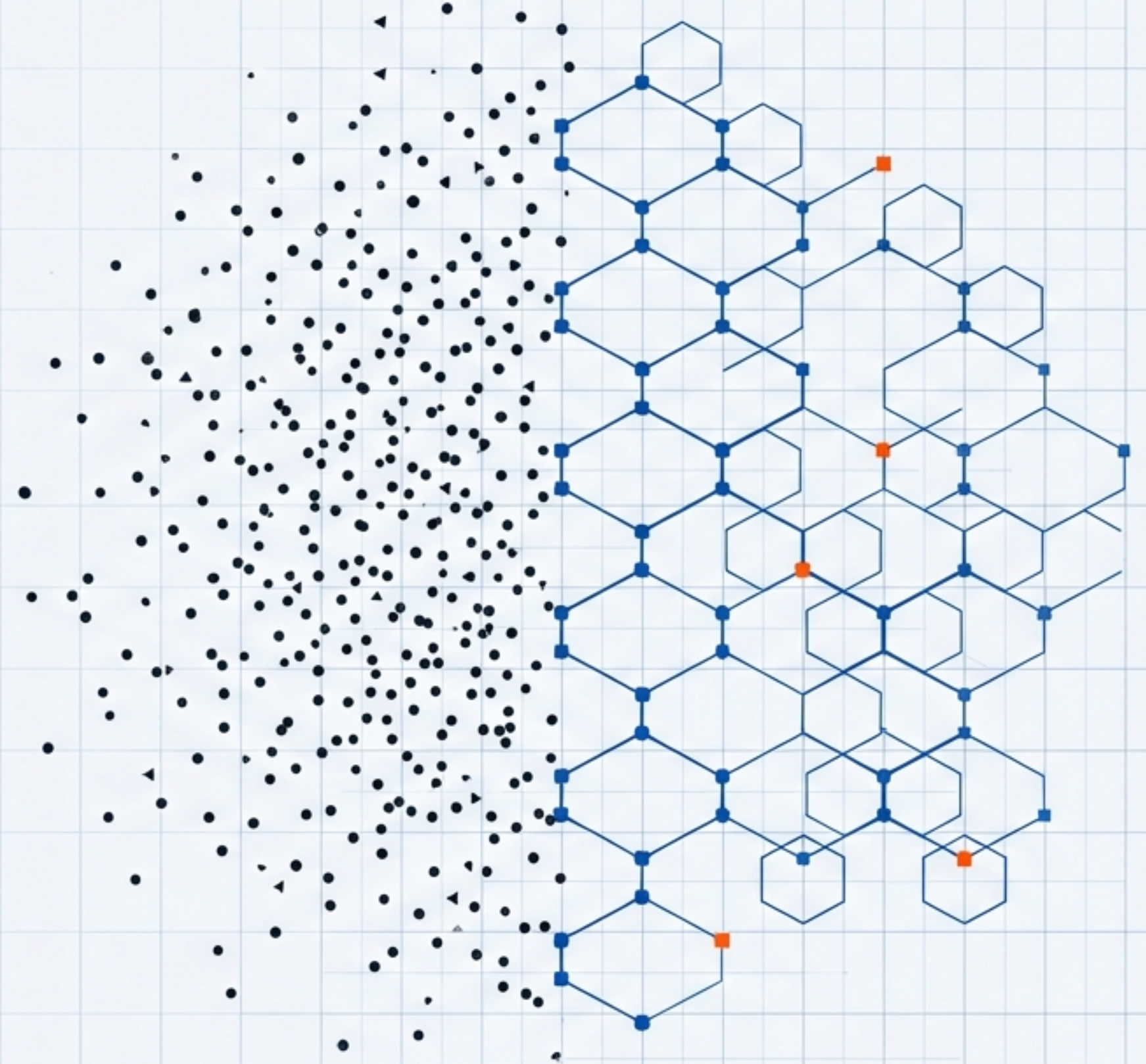


Repository-Level Ontology & Taxonomy

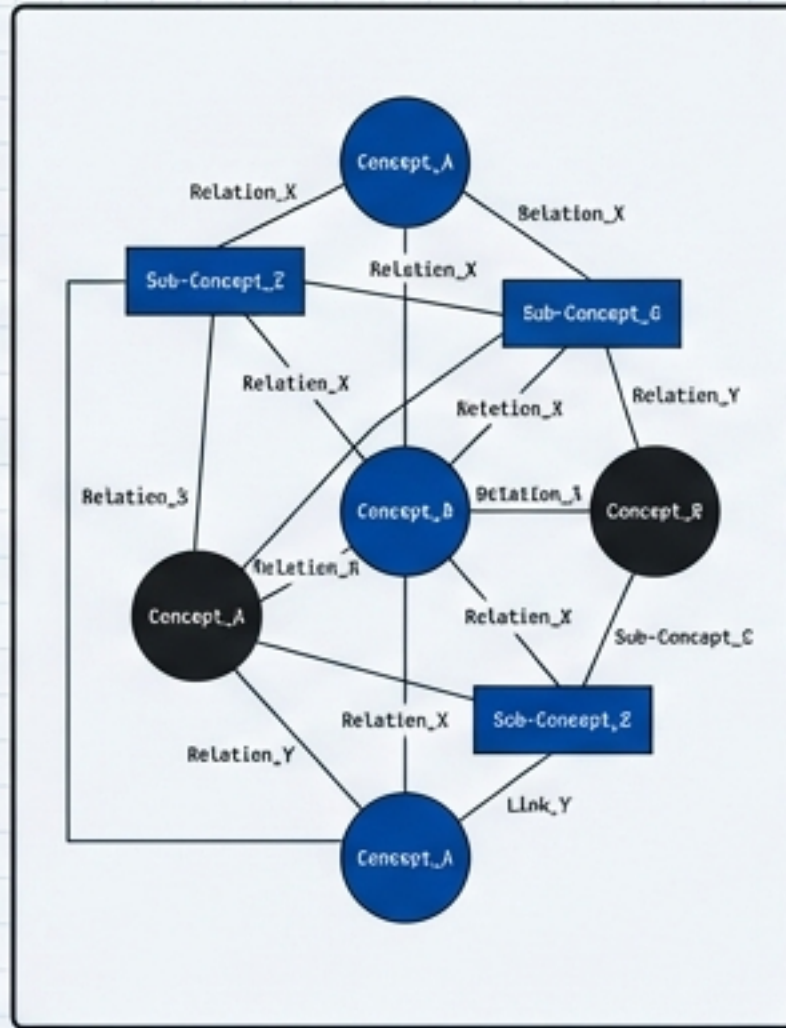
The Missing Layer—Applying Semantic
Knowledge Graph principles to the
repository itself.

DATE:	31 January 2026
AUTHOR:	Dinis Cruz
STATUS:	Design Phase
DEPENDENCY:	Three-Layer Content Architecture



We have mastered the micro, but missed the macro

Inside the File (Micro)



We create `SEMANTIC-GRAPH.md` files to map internal concepts and relations.

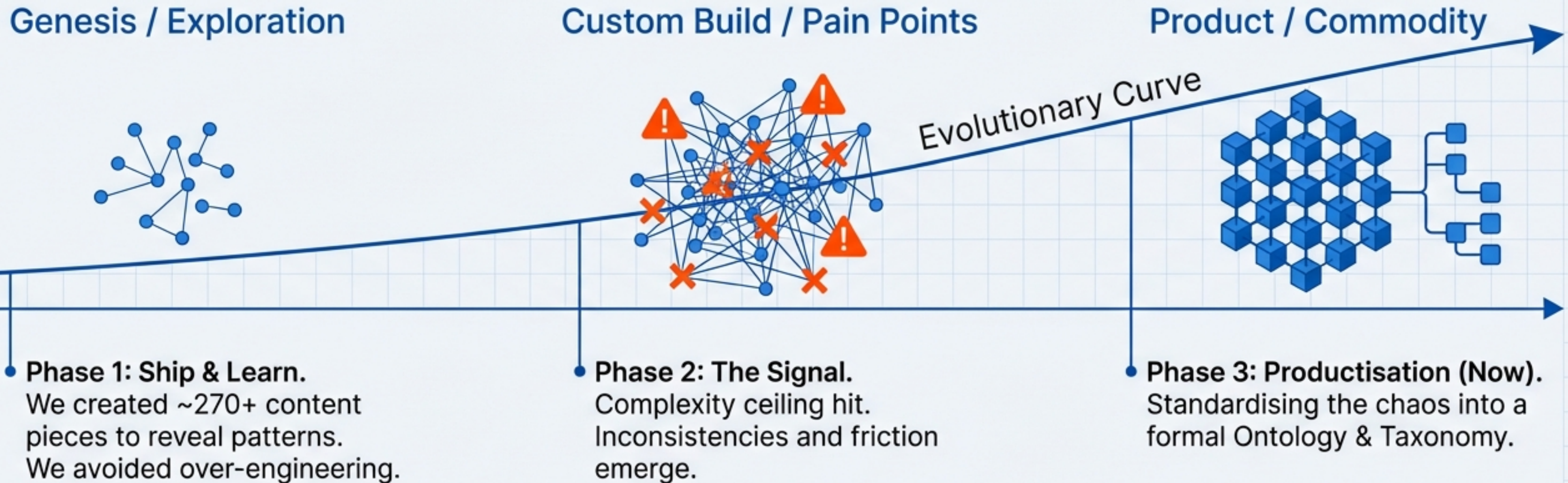
Outside the File (Macro)

```
root/  
  file_A.md  
  file_B.md  
  file_C.md  
  notes_2023.txt  
  report_final.pdf  
  project_data.json  
  archive.zip  
  summary.docx  
  config.yaml  
  readme.txt
```

The repository root is currently a flat, disconnected list. We lack an ontology for the collection itself.

Premature optimisation is the root of all evil

Strategic Context: The Wardley Mapping Doctrine

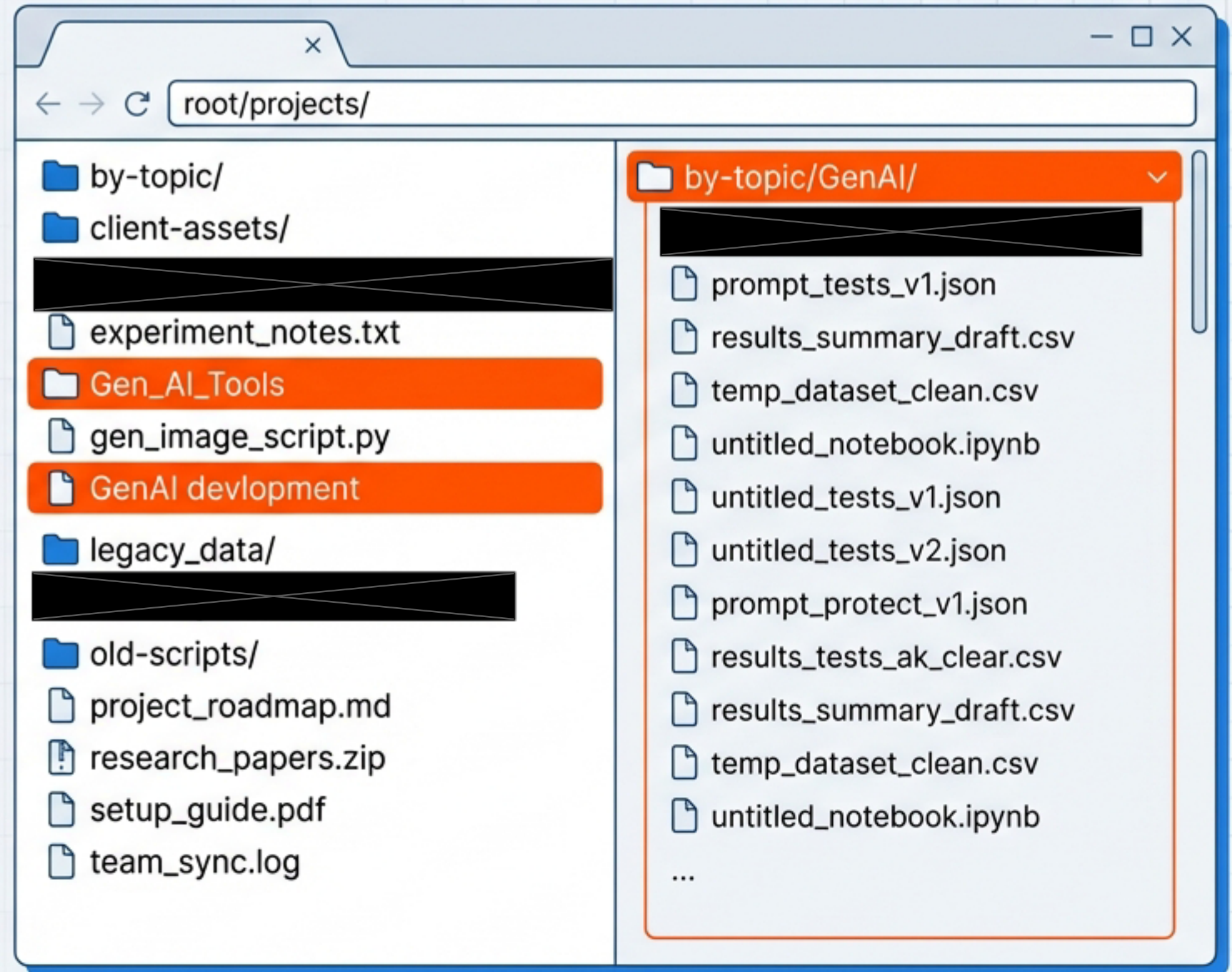


Wardley Doctrine: Don't try to create the perfect architecture before you understand the problem. Ship, learn, then improve.

The flat structure is breaking under its own weight

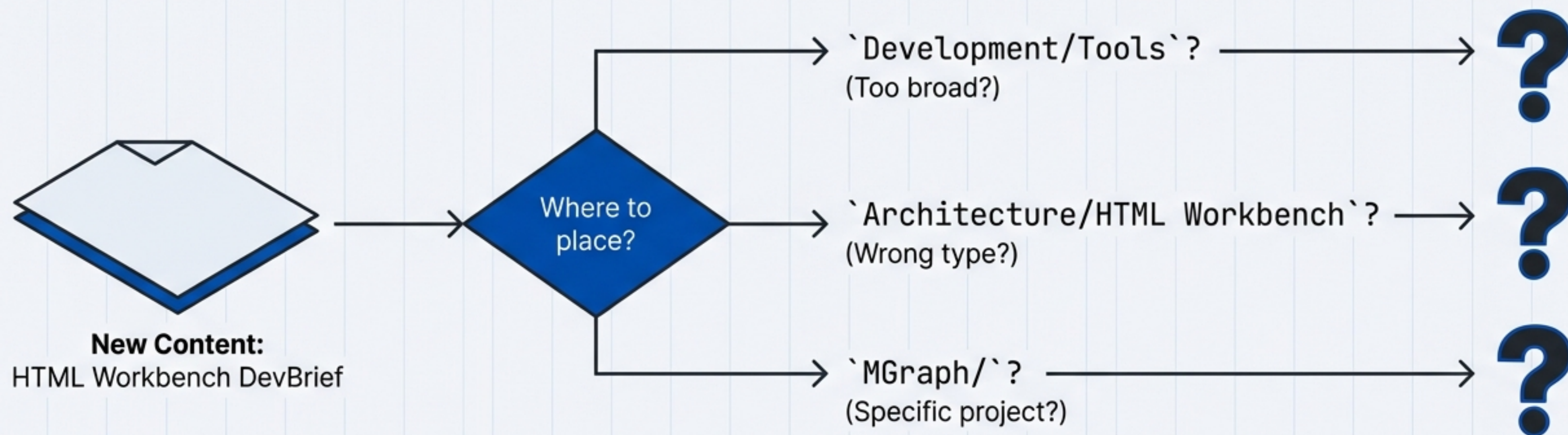
Current Structural Failures

- **Inconsistent Naming:** Search fails and cross-referencing becomes unreliable.
- **Persistent Typos:** unprofessional and hard to fix once paths are referenced.
- **Flat Scaling:** Folders overwhelm users with 20+ items and no sub-categorisation.



Every new file currently triggers a debate

The Ambiguity of Mixed Granularity



Real World Example

On Jan 29, the AI agent (Claude) paused processing to ask: *"Is DevBrief a folder or a tag? Should this go in Architecture or Tools?"* This manual bottleneck prevents automation.

Key Insight

Confusion source: We are mixing **'Content Type' (Format)** with **'Topic' (Subject)** in the same folder hierarchy.

Establishing the Semantic Layer



ONTOLOGY.md

The Schema.

Question: What types of things exist?

—

Role: Defines the “Content Types” and valid entities.



TAXONOMY.md

The Map.

Question: How are categories hierarchically structured?

—

Role: Defines the “Folder Structure” and parent/child relationships.

Together, these files create the logic for `semantic-data/`.

Separating Topic from Content Type

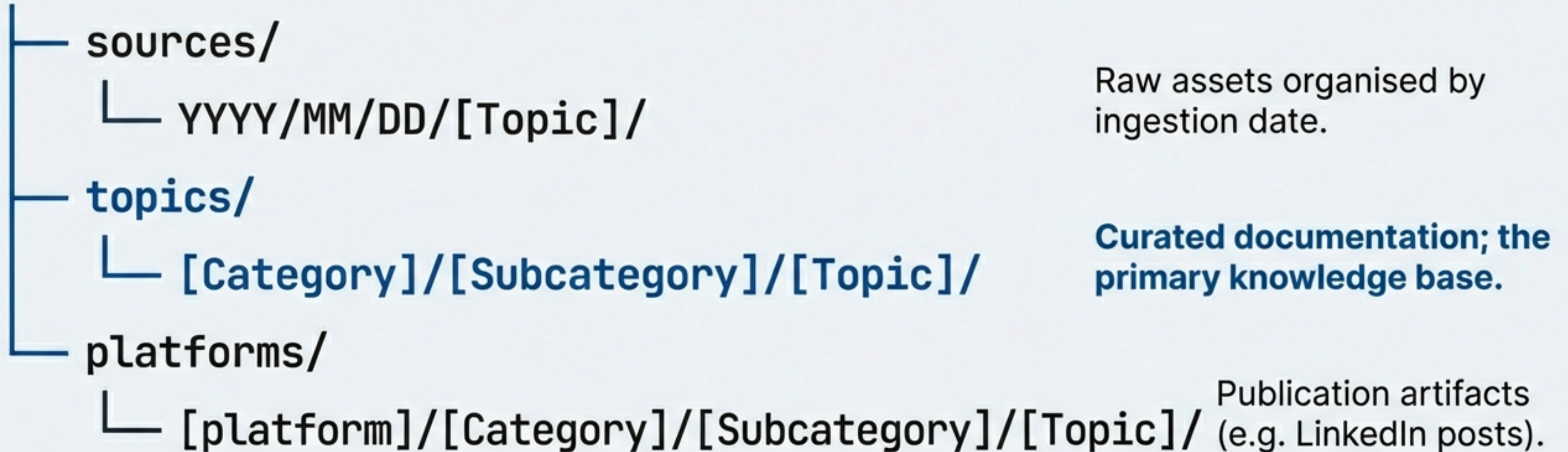
The Core Shift: Orthogonality

	DevBrief	Architecture	Solution	Research
HTML Workbench	✓	✓	Coming Soon	
Graph-Based Issues	✓		✓	
Three-Layer Architecture	✓			

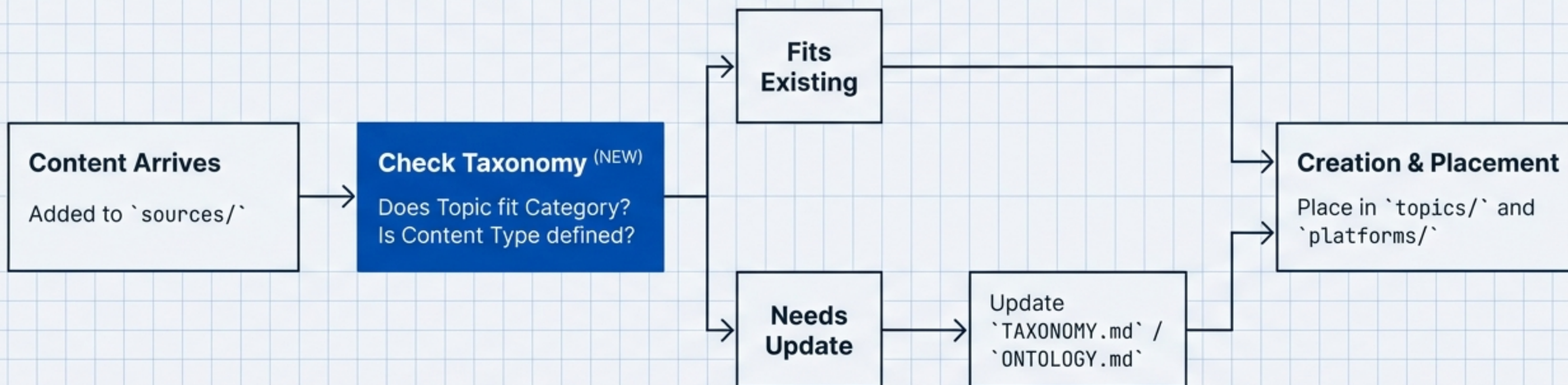
→ A single Topic can possess multiple Content Types. We no longer have to force a file into a single folder based on its format.

The New Directory Architecture

semantic-data/



Logic Flow: From Content to Placement



The Taxonomy in Practice

Category: GenAI

AI-powered development practices

- 📁 Claude Skills
- 📁 Code Quality
- 📁 LLM Workflows
- 📁 Vibe Coding

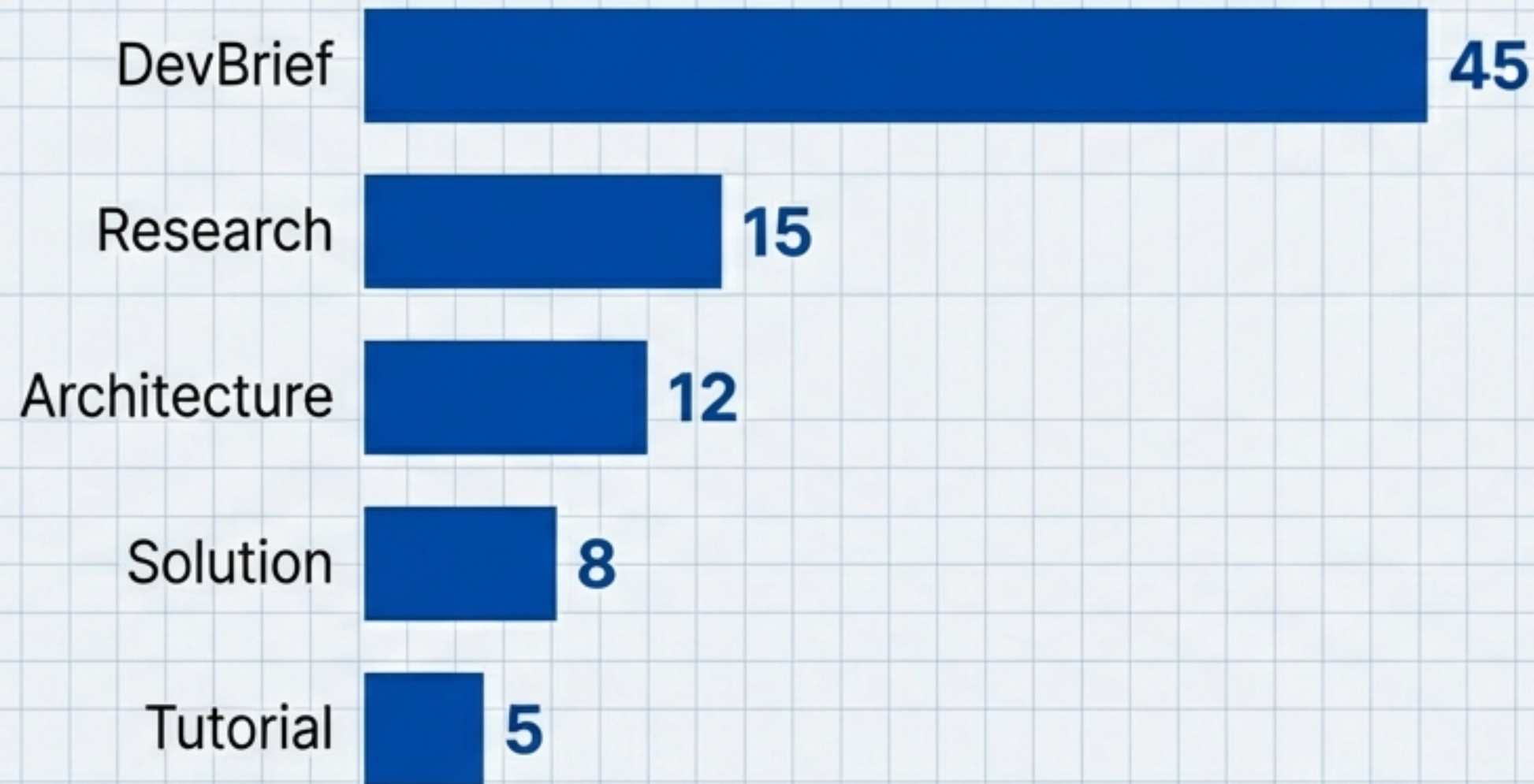
Category: Cybersecurity

Security, privacy, and compliance

- 📁 GDPR
- 📁 Privacy Architecture

📌 **Hierarchy Rule:** Defined subcategories **activate** when items exceed ~15-20 per folder.

Standardising Content Types (The Ontology)



DevBrief: Technical briefs and overviews

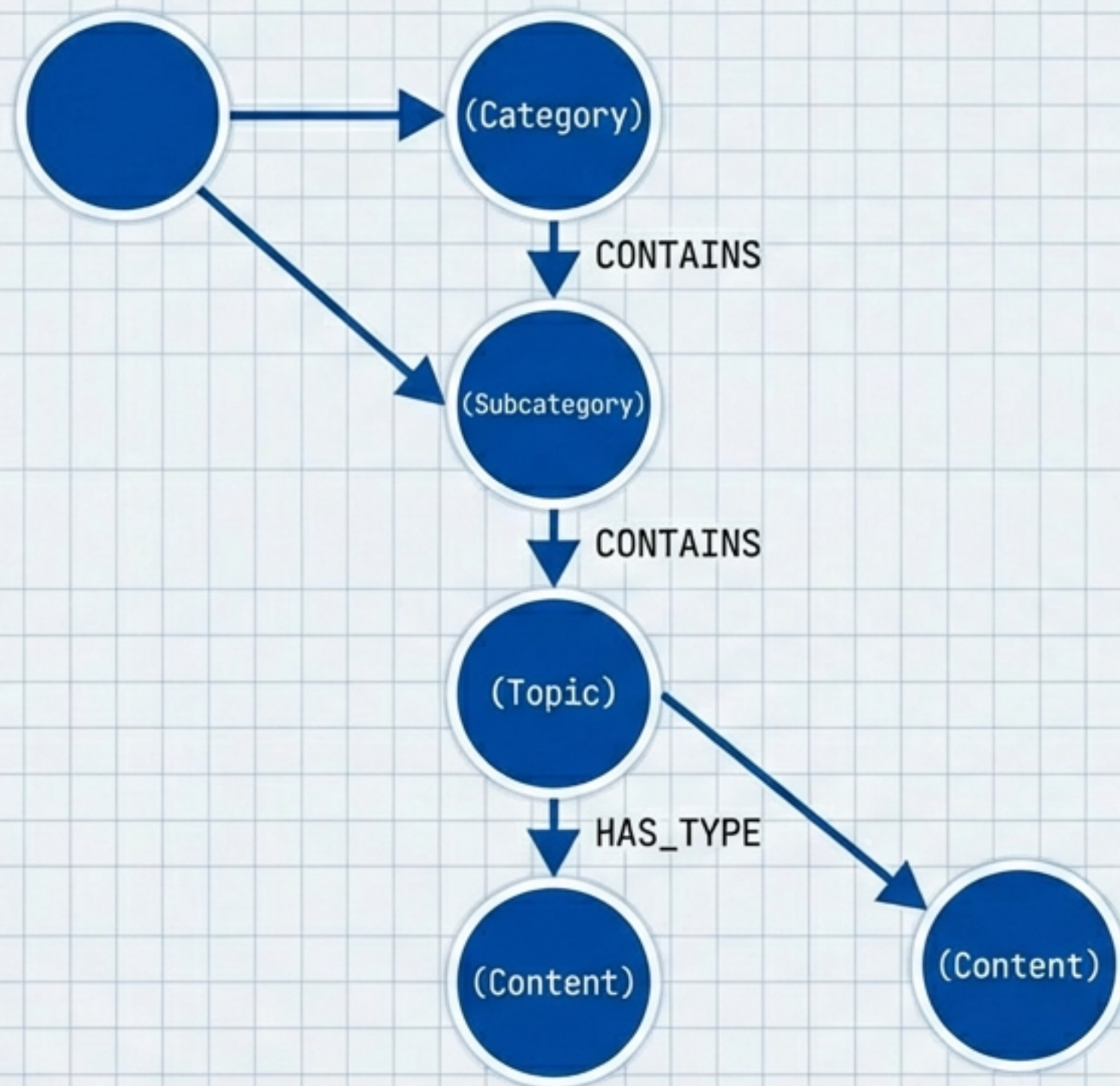
Architecture: System design documents

Solution: Working implementations

Research: Deep research reports

Tutorial: How-to guides

Built for Machines: Graph Integration



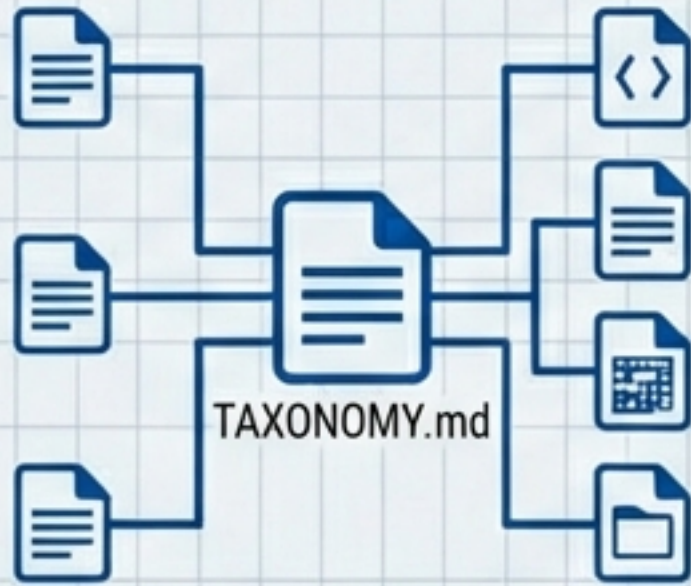
Cypher Schema:

The directory structure maps directly to Neo4j nodes.

This enables complex querying:

“Find all Architecture documents related to GenAI Security”.

The ROI of Structure



Single Source of Truth

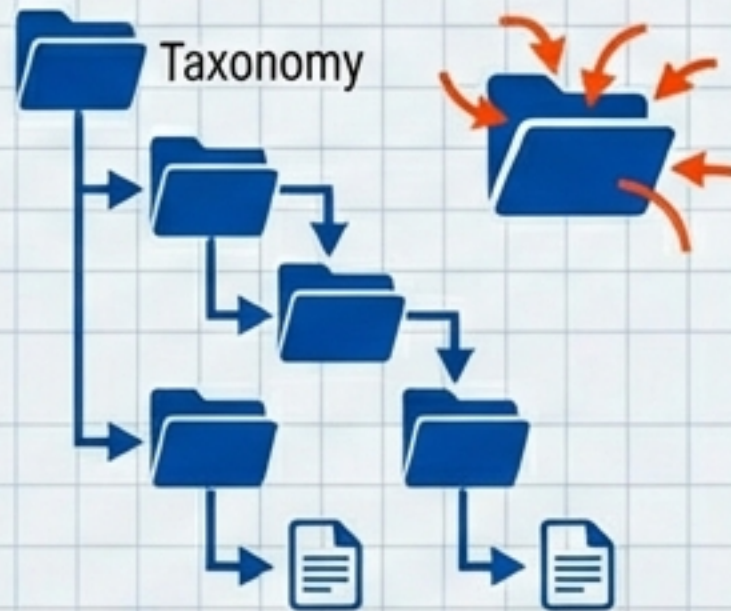
Category names defined once in `TAXONOMY.md`, used everywhere. Eliminates naming debates.



Professional Consistency

Eliminates typo propagation.

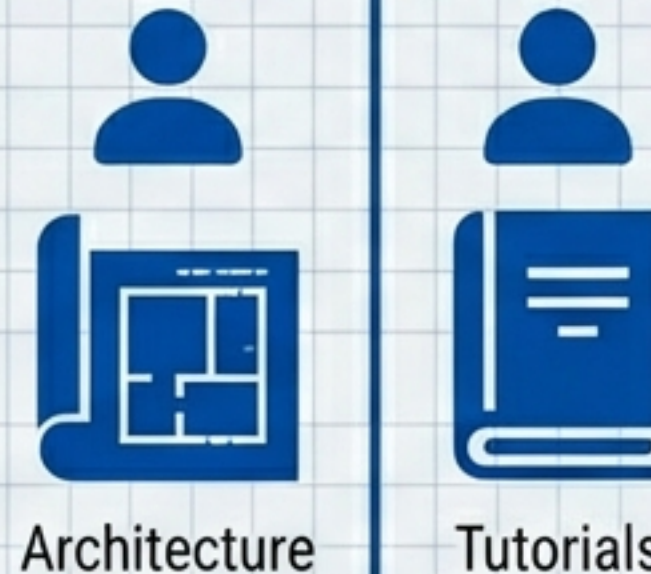
Ensures standard formatting across all 270+ files.



Scalable Navigation

Subcategories prevent folder overload.

The Taxonomy *is* the navigation map.



Clear Mental Models

Users know exactly where to look.

Distinguishes 'Architecture' from 'Tutorials' instantly.

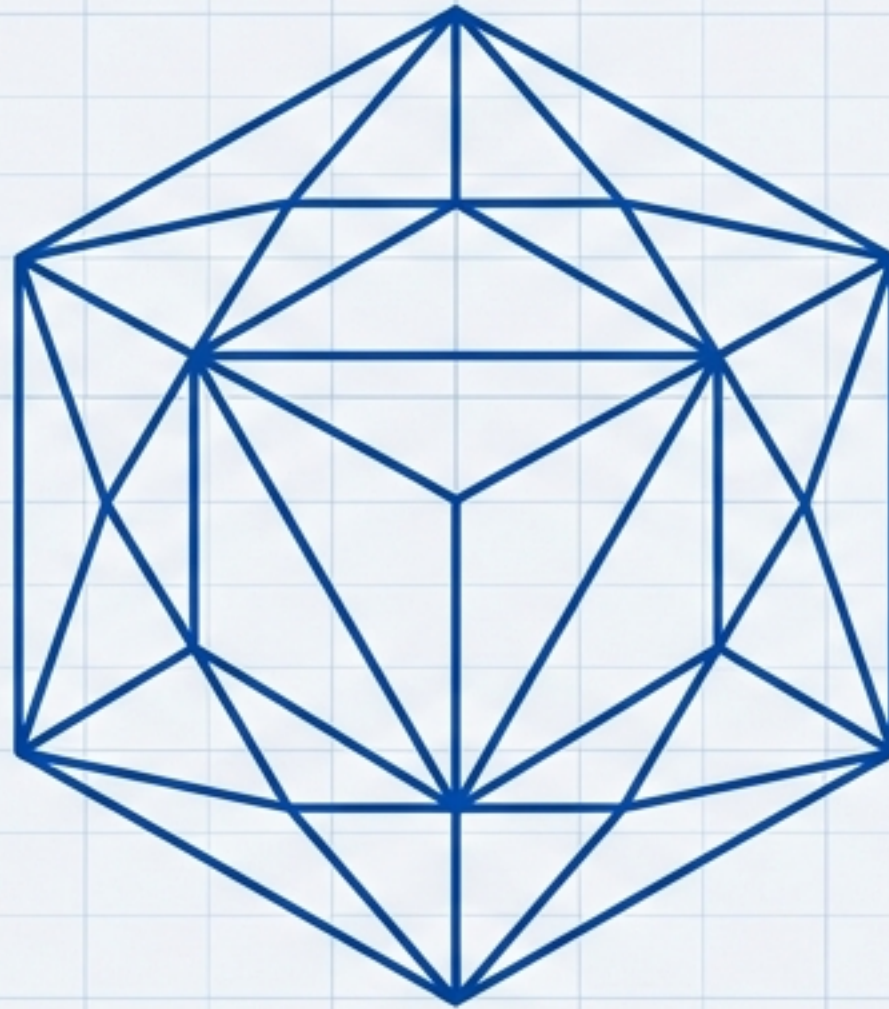
Migration & Implementation Plan

- ✓ Create ``semantic-data/`` directory structure.
- ✓ Refactor existing content (Starting with Jan 29 batch).
- ✓ Update ``.claude/`` workflow guides to use new Ontology logic.
- **Maintain Legacy Paths** (``by-topic``, ``curated-guides``) temporarily.

Migration Strategy: **Non-destructive**. Legacy paths remain active until refactoring is complete.

From Collection to Knowledge Base

Transforming a flat list of files into a semantic engine.



Action: Approve schema and commence migration to `semantic-data/`.